CSA1306

Theory of Computation.

CO-2

AT-1

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(2)

Task 1: Regular Expressions

1. Binary string ending with 01

$(0+1)^* 01$

2. Binary strings containing 11 or 00

$(0+1)^*(11+00)(0+1)^*$

3. string containing at least one ab

$(a+b)^* ab(a+b)^*$

4. String having an even numbers of zeros

$1^*(01^*01^*)^*$

5. strings starting with a and ending with b

$a(a+b)^* b$

Task 3: Regular Expression Analysis (or) context-free Grammar

3. Arithmetic expression with additio

Start Symbol: E

Terminals: a, +, b

Non-terminals: E

Production Rule:

$$E \rightarrow E+E \mid a \mid b$$

4. Balanced Parentheses

* Start Symbol: S
* Terminals: (,)
* Non-terminals: S

Production rules

$$S \rightarrow (S)\,S \mid \epsilon$$

5. Binary strings with equal zeros and ones

* Start symbol: S
* Terminals: 0, 1
* Non-terminals: S

Production rules:

$$S \rightarrow 0S1S \mid 1S0S \mid \epsilon$$

Task 4: Derivation and Parse Tree

The grammar is missing in assuming the inteded frammca is

$S \rightarrow aSb \mid \epsilon$

Generated strings

n	string
0	ε
1	ab
2	aabb
3	aaabbb

LMD:

$S \Rightarrow aSb$

$\rightarrow aaSbb$

$\Rightarrow aaaSbb$

$\Rightarrow aaabbb$

LMD

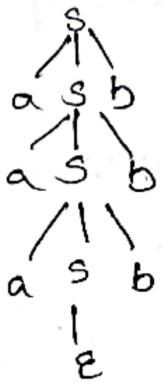
Parse tree:

$S \Rightarrow aSb$

$\Rightarrow aaSbb$

$\Rightarrow aaaSbbb$

$\Rightarrow aaabbb$



Task 5: Arithmetic Expression Grammar

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid id$

Derivation of a+b*c

$E \Rightarrow E+T$

$\Rightarrow T+T$

$\Rightarrow F+T$

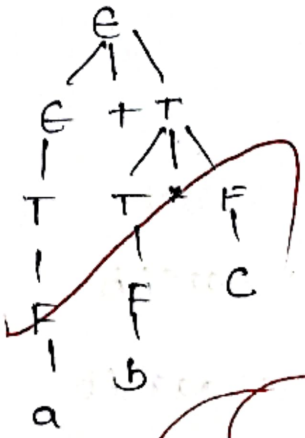
$\Rightarrow a+T$

$\Rightarrow a+T*F$

$\Rightarrow a+F*F$

$\Rightarrow a+b*c$

Parse Tree



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